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Chapter 1 Groups and Homomorphisms

Exercise 1

Let N be a subgroup of G and $g \in G$. Then $g^{-1}Ng = \{g^{-1}ng \mid n \in N\} = \{g^{-1}gn \mid n \in N\} = \{n \mid n \in N\} = N$ because G is abelian. So any subgroup of G is normal. Since G is simple, the only subgroups of G are $\{1\}$ and G . Also, using the fact that G is simple we can assume that $G \neq \{1\}$. Choose $h \in G \setminus \{1\}$. Since $\langle h \rangle$ is a subgroup of G , $\langle h \rangle = G$ or $\langle h \rangle = \{1\}$. If $\langle h \rangle = \{1\}$, then $h = 1$, a contradiction. Hence, $\langle h \rangle = G$, so G is a cyclic group. Let $|G| = n$. Assume for contradiction that n is not prime, so $n = ab$ for some $1 < a, b < n$. Since G is cyclic and a is a factor of n , there exists $k \in G$ of order a . Since $|k| > 1$, $k \neq 1$. By the above argument, we have $\langle k \rangle = G$. Therefore, $a = |k| = |\langle k \rangle| = |G| = n$, a contradiction. Hence, G is cyclic of prime order. $\square$

Exercise 2

Assume $H \neq \{1\}$. Let $g_1, g_2 \in G$. Suppose $g_1\vartheta = g_2\vartheta$. Then $(g_1\vartheta)(g_2\vartheta)^{-1} = 1$ and thus $(g_1g_2^{-1})\vartheta = 1$. Hence $g_1g_2^{-1} \in \text{Ker}(\vartheta)$. Since $\text{Ker}(\vartheta) \triangleleft G$, we have $\text{Ker}(\vartheta) = \{1\}$ or $\text{Ker}(\vartheta) = G$. If $\text{Ker}(\vartheta) = G$, then $g\vartheta = 1$ for all $g \in G$. Since ϑ is surjective, $H = \text{Im}(\vartheta) = \{g\vartheta \mid g \in G\} = \{1\}$, a contradiction. Therefore, $\text{Ker}(\vartheta) = \{1\}$. Hence, we have $g_1g_2^{-1} = 1$, which implies $g_1 = g_2$. So ϑ is injective and thus is bijective. $\square$

Chapter 2 Vector Spaces and Linear Transformations

Exercise 3

Suppose $V = U \oplus W$. Clearly, $V = U + W$. Let $v \in U \cap W \subseteq V$. Then there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. So $u + w \in U$ and $u + w \in W$. By the uniqueness of u and w , we have $u + w = u$ and $u + w = w$, so $u = w = 0$, and thus $v = 0$. Hence, $U \cap W = \{0\}$.

Now suppose $V = U + W$ and $U \cap W = \{0\}$. We want to prove that for any $v \in V$ there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. Let $v \in V$. Suppose $u_1, u_2 \in U$ and $w_1, w_2 \in W$ satisfy $v = u_1 + w_1 = u_2 + w_2$. Then $u_1 - u_2 = w_2 - w_1$. Since U and W are subspaces, they are closed under vector addition and additive inverses, so $u_1 - u_2 \in U$ and $w_2 - w_1 \in W$, so $u_1 - u_2 = w_2 - w_1 \in U \cap W$. Hence, $u_1 - u_2 = w_2 - w_1 = 0$, and thus $u_1 = u_2$ and $w_1 = w_2$, proving the uniqueness of u and w .

Therefore, $V = U \oplus W$ if and only if $V = U + W$ and $U \cap W = \{0\}$. $\square$

Exercise 4

Suppose $V = U \oplus W$. Let $v \in V$. Then there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. Since $u_1, \dots, u_r$ is a basis of U , there exist unique $\lambda_1, \dots, \lambda_r \in F$ such that $u = \lambda_1 u_1 + \dots + \lambda_r u_r$. Since $w_1, \dots, w_s$ is a basis of W , there exist unique $\mu_1, \dots, \mu_s \in F$ such that $w = \mu_1 w_1 + \dots + \mu_s w_s$. Therefore, there exist unique $\lambda_1, \dots, \lambda_r, \mu_1, \dots, \mu_s \in F$ such that $v = \lambda_1 u_1 + \dots + \lambda_r u_r + \mu_1 w_1 + \dots + \mu_s w_s$. Hence, $u_1, \dots, u_r, w_1, \dots, w_s$ is a basis of V .

Now suppose $u_1, \dots, u_r, w_1, \dots, w_s$ is a basis of V . Let $v \in V$. Then there exist unique $\lambda_1, \dots, \lambda_r, \mu_1, \dots, \mu_s \in F$ such that $v = \lambda_1 u_1 + \dots + \lambda_r u_r + \mu_1 w_1 + \dots + \mu_s w_s$. Since $u_1, \dots, u_r$ is a basis of U , there exists a unique $u \in U$ such that $u = \lambda_1 u_1 + \dots + \lambda_r u_r$. Since $w_1, \dots, w_s$ is a basis of W , there exists a unique $w \in W$ such that $w = \mu_1 w_1 + \dots + \mu_s w_s$. Hence, there exist unique $u \in U$ and $w \in W$ such that $v = u + w$. So $V = U \oplus W$.

Therefore, $V = U \oplus W$ if and only if $u_1, \dots, u_r, w_1, \dots, w_s$ is a basis of V . $\square$

Chapter 3 Group Representations

Exercise 1

Suppose ρ is a representation of G over $\mathbb{C}$. Then

$$\begin{aligned} A^m &= (a\rho)^m \\ &= a^m \rho \\ &= 1\rho \\ &= I. \end{aligned}$$

Now suppose $A^m = I$. Let $a^r, a^s \in G$, where $0 \leq r, s \leq m-1$. If $r+s \leq m-1$, then

$$\begin{aligned} (a^r a^s)\rho &= a^{r+s}\rho \\ &= A^{r+s} \\ &= A^r A^s \\ &= (a^r)\rho(a^s)\rho. \end{aligned}$$

If $m \leq r + s \leq 2m - 2$, then

$$\begin{aligned}
(a^r a^s)\rho &= a^{r+s}\rho \\
&= a^{r+s-m}\rho \\
&= A^{r+s-m} && \text{(note that } 0 \leq r + s - m \leq m - 2 \leq m - 1\text{)} \\
&= A^r A^s A^{-m} \\
&= A^r A^s (A^m)^{-1} \\
&= (a^r)\rho(a^s)\rho I^{-1} \\
&= (a^r)\rho(a^s)\rho.
\end{aligned}$$

Therefore, ρ is a homomorphism, and hence a representation. $\square$

Exercise 4

Let $g \in G$. Note that $g\rho = I^{-1}(g\rho)I$ and I is invertible. So ρ is equivalent to ρ , proving (1).

Suppose ρ is equivalent to σ , so we have $g\sigma = T^{-1}(g\rho)T$ for some invertible matrix T . Then $g\rho = (T^{-1})^{-1}(g\sigma)T^{-1}$ and T^{-1} is invertible, so σ is equivalent to ρ , proving (2).

Now suppose ρ is equivalent to σ and σ is equivalent to τ . We have $g\sigma = T^{-1}(g\rho)T$ and $g\tau = S^{-1}(g\sigma)S$ for some invertible matrices T and S . Then $g\tau = S^{-1}T^{-1}(g\rho)TS = (TS)^{-1}(g\rho)TS$ and TS is invertible, so ρ is equivalent to τ , proving (3). $\square$

Exercise 6

Let

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

and

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Since $A^4 = B^2 = I$ and $B^{-1}AB = A^{-1}$, by Example 1.4, $\rho : D_8 \rightarrow \text{GL}(3, \mathbb{R})$ defined by $(a^i b^j)\rho = A^i B^j$ is a homomorphism. So ρ is a representation of D_8 of degree 3.

We can use the following Python code to compute the kernel of ρ .

```

1  import sympy as sp
2
3  A = sp.Matrix([
4      [0, 1, 0],
5      [-1, 0, 0],
6      [0, 0, 1]
7  ])
8
9  B = sp.Matrix([
10     [1, 0, 0],
11     [0, -1, 0],
12     [0, 0, 1]
13 ])

```

```

14
15 print("Elements in the kernel ((i, j) means a^i b^j):")
16 for i in range(4):
17     for j in range(2):
18         if (A**i) * (B**j) == sp.eye(3):
19             print(f"({i}, {j})")

```

We get $\text{Ker}(\rho) = \{a^0b^0\} = \{1\}$. Therefore, ρ is faithful. $\square$

Exercise 7

By the first isomorphism theorem, $G/\text{Ker}(\rho) \cong \text{Im}(\rho) \leq \text{GL}(1, F) \cong F^\times$. Since $F^\times$ is abelian, $G/\text{Ker}(\rho)$ is also abelian. $\square$

Exercise 8

The statement is false. A counter-example is the trivial representation of a non-abelian group. $\square$